DESIGN MILESTONES

Coffee Machine — Beverage Decision Fatigue

IDE302 Design Studio · Sude Söylemez S040431 · 2025–2026 Spring

AI ROLE TAGS

Clarifier
Structures raw research into insight

Expander
Opens new directions & opportunities

Re-framer
Shifts the entire problem frame

Critique
Evaluates & pushes refinement

DESIGN PROCESS TIMELINE

- WK 2 · Start
- WK 3A · Define
- WK 3B · Pivot
- WK 4 · Concept
- WK 5 · Proto
- WK 6 · Test

01

WEEK 2

Feb 15–19, 2026

Empathize & Define

CLARIFIER

#kettle-interaction #user-needs #usability #safety #control-mechanisms

SELECTED PROMPT

"What are the typical needs, annoyances, and expectations of users connected to regular use of an electric kettle? Pay attention to everyday interaction, safety, and usability."

AI RESPONSE SUMMARY

Clarifier

Identified core user needs: fast boiling, controlled pouring, clear feedback, temperature precision. Separated frustrations (grip, cleaning difficulty, poor feedback) from actual needs. Flagged interaction logic as the key under-explored design space.

REFLECTION

AI helped me stop assuming and start seeing patterns. Shifting focus from physical form to interaction logic felt like the real discovery — the kettle was never just a heating device.

DESIGN DECISION

Rethink kettle as an interaction product. Focus on adaptable control and feedback clarity.

✓ ACCEPTED

User control & temperature precision as core design directions

✗ REJECTED

Purely technical / hardware-focused solutions (faster boiling, larger tank)

⟳ TRANSFORMED

Form exploration → Interaction logic & control mechanism design

02

WEEK 3

Feb 22, 2026

Define — Opportunity Mapping

EXPANDER

#opportunity-areas #experience-orientation #beverage-ritual #interaction-clarity

SELECTED PROMPT

"Based on common kettle user needs and frustrations, what potential opportunity areas can be explored for improving interaction and user control?"

AI RESPONSE SUMMARY

Expander

Proposed reframing the kettle as an experience-oriented product. Key opportunity areas: temperature control UX, feedback clarity, interaction simplification, and beverage ritual support. Positioned 'ritual' as a design lens.

REFLECTION

AI shifted my thinking from fixing frustrations to opening possibilities. The 'beverage ritual' framing changed how I saw the entire product — from a utility tool to a moment of intentional pause.

DESIGN DECISION

Pivot focus toward interaction experience over technical specs. Adaptability as key value.

✓ ACCEPTED

Experience-oriented framing; interaction simplification as the primary design goal

✗ REJECTED

Tech-heavy, feature-stacking approaches to problem solving

⟳ TRANSFORMED

User frustrations → Design opportunity areas framed as experience gaps

03

WEEK 3

Feb 25, 2026

Pivot — Beverage Decision Fatigue

RE-FRAMER

#decision-fatigue #cognitive-load #psychological-tension #mental-relief #ritual

SELECTED PROMPT

"What psychological tensions exist in everyday beverage selection behaviors, particularly under cognitive fatigue conditions? Write a short and clear answer."

AI RESPONSE SUMMARY

Re-framer

Four core tensions: (1) Mental relief ↔ Need to decide, (2) Control desire ↔ Decision avoidance, (3) Functional choice ↔ Emotional intention, (4) Efficiency ↔ Ritualistic meaning. Core insight: users aren't choosing a drink — they're negotiating psychological state.

REFLECTION

This was the pivotal moment of the entire project. AI reframed everything. The problem was never about kettles — it was about the psychological burden of micro-decisions during mental recovery moments.

DESIGN DECISION

Full concept pivot. Reduce decision fatigue without removing agency. Concept 01 selected: State-Responsive Guidance System.

✓ ACCEPTED

Decision fatigue as core design tension. Mental Relief ↔ Need to Decide as guiding frame

✗ REJECTED

Concept 03 — Decision-Free Default Flow (strips all ritual meaning, too automated)

⟳ TRANSFORMED

Kettle redesign → Experience design for psychological regulation rituals

04

WEEK 4

Mar 01–04, 2026

Concept Development

EXPANDER

#dual-bean #recipe-logic #guided-base #progressive-narrowing #caffeine-control

SELECTED PROMPT

"I want coffee-based beverage recipes. User creates the coffee base with the machine but chooses other ingredients. Horizontally designed, minimalist-industrial look. Include both decaf and caffeinated options."

AI RESPONSE SUMMARY

Expander

Proposed 'Guided Coffee Base System': machine automates coffee base only; user selects recipe and completes it. Uses progressive narrowing (category → 3 options → 1 selection). Step-by-step guidance keeps user involved without cognitive overload.

REFLECTION

AI transformed a vague direction into a concrete product concept. The Thermomix analogy clarified the interaction logic immediately. I also realized the physical form had to follow the concept — dual chambers make the idea visible.

DESIGN DECISION

Final concept: Recipe-Guided Coffee Machine. Machine brews base; user completes drink. Over-automation deliberately removed.

✓ ACCEPTED

Dual-bean hoppers (caffeinated + decaf), recipe-guided interaction, step-by-step guided flow

✗ REJECTED

Mood-based selection logic (caffeine ≠ mood state), built-in syrup modules, single-bean system

⟳ TRANSFORMED

Mood-based automation → Recipe logic with visible user participation

05

WEEK 5

Mar 04–08, 2026

Prototype & Interface Refinement

CRITIQUE

#interface-architecture #flavor-fate #three-path-system #UX-flow #visual-prototype

SELECTED PROMPT

"Organize the Coffee Mania interface. Users should choose from existing coffee types, select espresso-based recipes, and get beverage suggestions. Main feature: suggestion section — users define calm/wake-up/strong/frothy/milky preferences."

AI RESPONSE SUMMARY

Critique

Structured 3-path interface: (1) Coffee Menu — direct selection, (2) Personalised Beverage Suggestion — Flavor Fate quiz with sliders, (3) Recipe Lab — café-style recipes. Critique: simplify question flow, add visual icons, make interaction feel more game-like.

REFLECTION

Reducing menu options alone doesn't solve decision fatigue — the structure of choices matters more than the count. Mood-labelled buttons introduced new complexity. Short guided quiz with visual sliders proved far more effective.

DESIGN DECISION

Final interface: 3-path home screen. Beverage Suggestion as primary. No knobs — fully digital touchscreen. Planned: icon system, simplified quiz, game-like visual language.

✓ ACCEPTED

3-path architecture, dual-bean visibility on home screen, step-by-step recipe flow, Decaf/Caffeinated toggle

✗ REJECTED

Knob-based physical controls, over-long preference tests, mood-labelled category buttons

⟳ TRANSFORMED

12-drink flat menu → Progressive narrowing: mood → frothiness → flavor → 1 recommendation

06

WEEK 6

Mar 11, 2026

Test / Communicate — Bean Logic Redesign

EXPANDER

#bean-profiles #flavor-logic #custom-bean #interface-iteration #recommendation-algorithm

SELECTED PROMPT

"Generate visual mock-ups of bean selection screens (Step 1: bean profile selection and Step 2: custom bean description) that match the coffee machine's interface style. Explore alternative interaction structures and layout options."

AI RESPONSE SUMMARY

Expander

Generated visual mock-ups for a two-step bean selection interface: Step 1 — choose a bean profile (Soft & Smooth, Balanced, Bright, Rich, Bold, Decaf); Step 2 — optionally define custom attributes (roast level, acidity, body, bitterness). Accelerated layout iteration; final interaction logic and algorithm decisions were defined by the designer.

REFLECTION

Instructor feedback revealed that limiting chambers to caffeine level alone (decaf vs. caffeinated) constrained the system too much. People experience coffee through flavor, body, and intensity — not just caffeine. Expanding to flavor-based profiles allowed the recommendation engine to respond more intelligently. Designing the bean setup interface also clarified how the algorithm would interpret bean characteristics, improving coherence between the physical machine and the interface.

DESIGN DECISION

Bean chamber logic redesigned to support 6 flavor profiles + Custom Bean. Two-step interface: profile selection → custom attribute definition (roast, acidity, body, bitterness). Recommendation system now adapts to both user mood and bean flavor identity.

✓ ACCEPTED

Multi-profile bean system (6 profiles + Custom Bean). Two-step bean setup interface. Flavor-based recommendation logic.

✗ REJECTED

Strict decaf vs. caffeinated binary chamber logic (too limiting, prevents taste-based adaptation)

⟳ TRANSFORMED

Caffeine-only selection → Flavor-profile-based bean system enabling mood & taste-driven recommendations

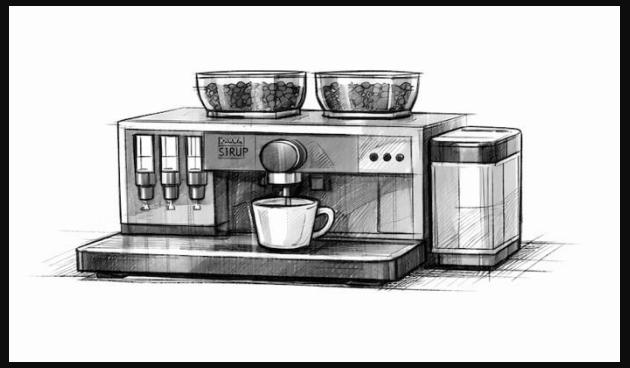
FINAL DIRECTION

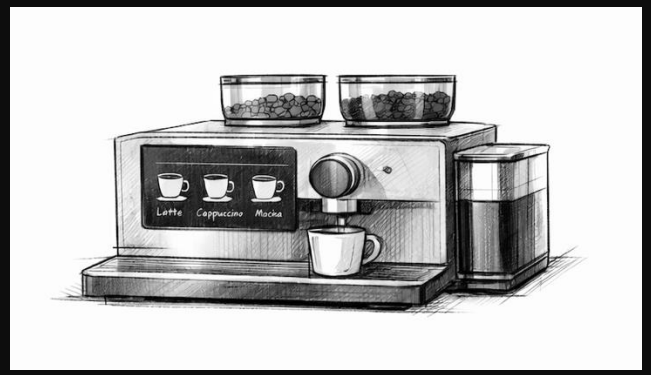
"A recipe-guided coffee machine with a flavor-profile-based bean system that reduces beverage decision frustration through a short, structured selection process and one optimized recommendation — adapting drink suggestions to both user mood and the loaded bean's flavor identity."

PROCESS SUMMARY · AI ROLE × DECISION OUTCOMES

WEEK	PHASE	AI ROLE	✓ ACCEPTED	✗ REJECTED	↻ TRANSFORMED
WK 2	Empathize & Define	Clarifier	Interaction logic & control	Hardware-only fixes	Form → Interaction design
WK 3A	Opportunity Mapping	Expander	Experience-oriented framing	Feature-stacking approaches	Frustrations → Experience gaps
WK 3B	Decision Fatigue Pivot	Re-framer	Mental Relief ↔ Need to Decide	Fully automated concept (C03)	Kettle → Psychological ritual
WK 4	Concept Development	Expander	Dual-bean, recipe-guided system	Mood-based logic / syrup modules	Automation → Guided participation
WK 5	Prototype & Refinement	Critique	3-path architecture, toggle system	Knob controls, long quizzes	Flat menu → Progressive narrowing
WK 6	Test / Bean Logic Redesign	Expander	6 flavor profiles + Custom Bean setup	Caffeine-only binary chamber logic	Caffeine logic → Flavor-profile recommendations

FINAL DIRECTION





FINAL PRODUCT

FINAL INTERFACE

"COFFEE BEANS" section was introduced to allow the machine to understand which coffee beans are currently loaded in the dual bean chambers. Since different beans have different flavor characteristics, the system needs this information to generate more accurate beverage recommendations. Instead of relying only on caffeine level, the machine can adjust drink suggestions, strength, and recipes based on the selected bean profile. This enables a more flexible and personalized beverage experience while maintaining the core goal of reducing decision fatigue.

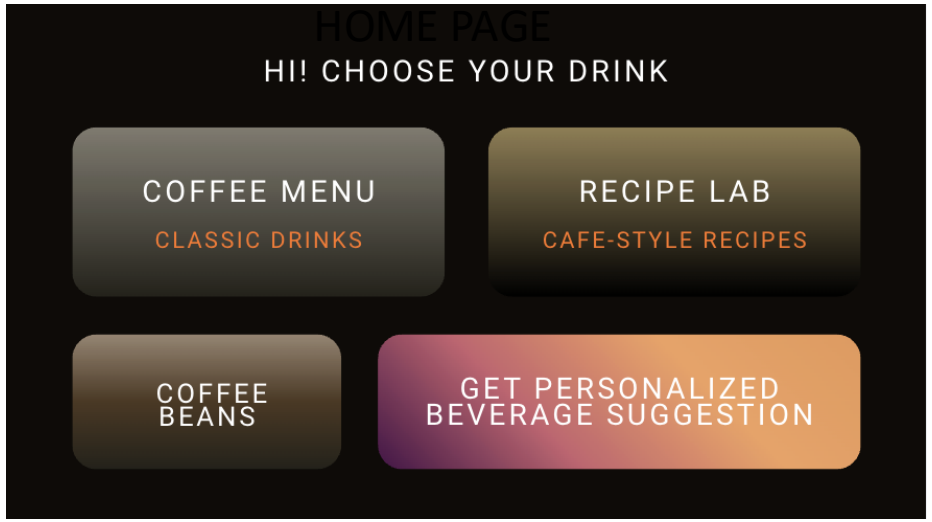
HOME PAGE

FINAL INTERFACE

“COFFEE MENU” page presents common espresso and milk based beverages that can be prepared by most standard espresso machines.

HOME PAGE

FINAL INTERFACE

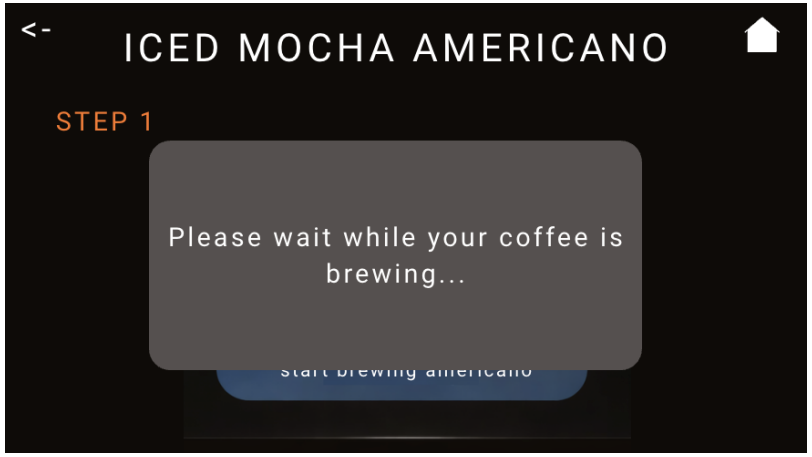


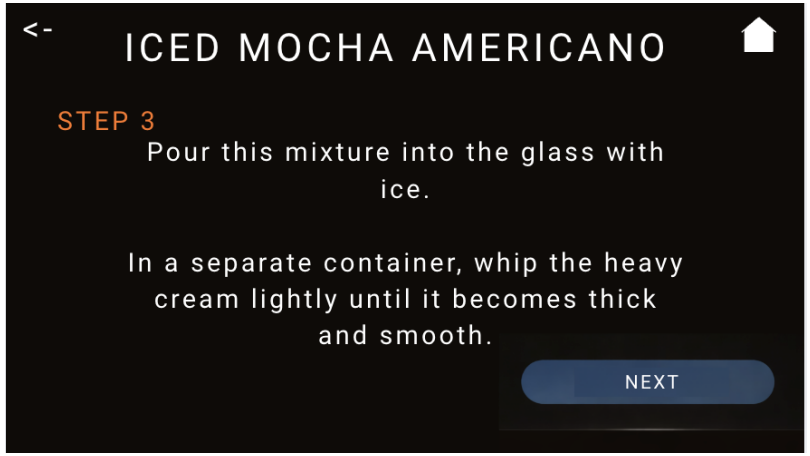
The “RECIPE LAB” section is one of the key features that differentiates this coffee machine from others. By offering a variety of suggested coffee recipes, it brings a café-style experience into the user’s home. The drinks recommended in this section are designed to be easily prepared with common household ingredients, while also including popular café-style beverages and additional creative variations.

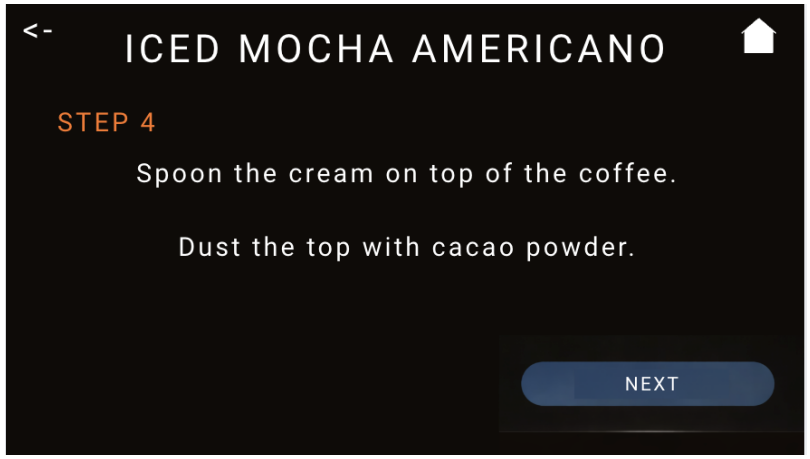
FINAL INTERFACE

When you select your drink, the algorithm first displays the preparation time and the required ingredients. It also allows you to choose which coffee bean chamber to use based on the beans currently loaded in the machine (decaf and caffeinated are only example options).

When you press “Start Recipe”, the step-by-step preparation of the selected drink appears on the screen, explaining the recipe in the simplest and clearest way so the user can easily follow the process.

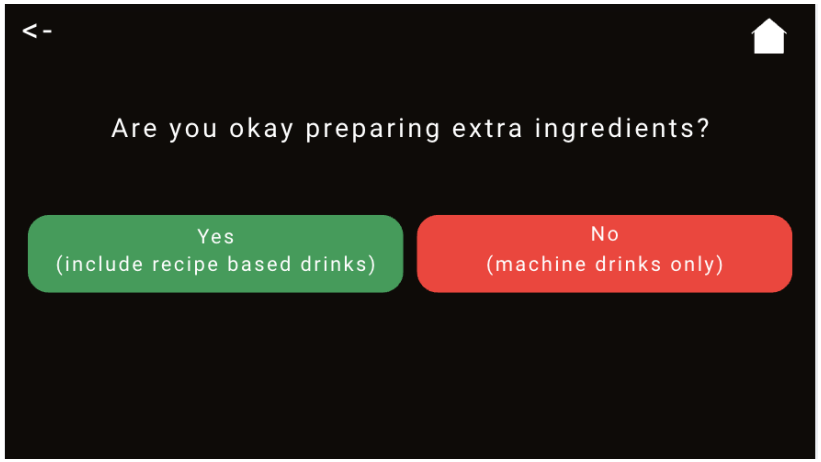
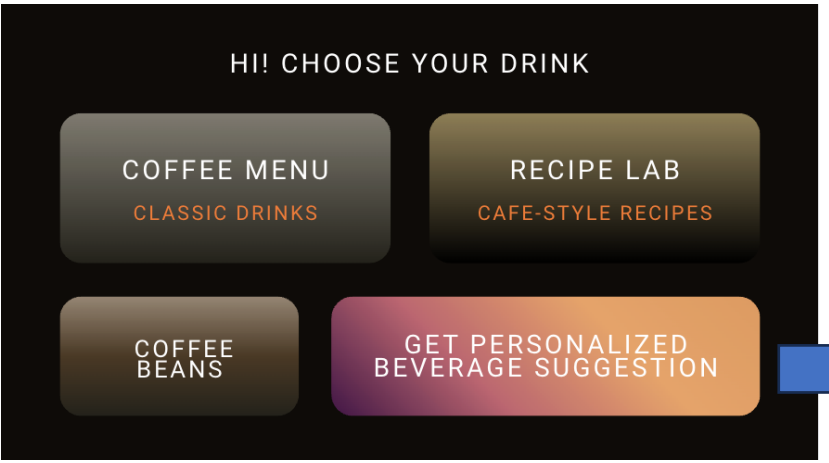






FINAL INTERFACE

The “Get Personalised Beverage Suggestion” section is the core feature of the product. This section was designed to eliminate the difficulty of choosing between a wide variety of beverage options. To better understand the user’s current preference, the machine asks a few short questions about the type of drink they want at that moment and then recommends a suitable beverage. The system also analyzes the coffee beans available in the machine’s chambers based on factors such as caffeine level and flavor profile, and suggests the most appropriate bean for the recommended drink. When the user presses **Start Recipe**, the preparation process begins with step-by-step instructions. If the user does not like the recommendation, the system can also suggest alternative drinks that match their preferences.



HOME PAGE